INJECTIVITY OF THE QUOTIENT-SHEAF MAP UNDER RESTRICTING A PRE-EQUIVALENCE RELATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-quotient-pre-equivalence (canonical tag 045Y). cited at proof lines 4060-4062. Pointwise description of the quotient sheaf.
- Lemma sites-lemma-covering-surjective-after-sheafification (canonical tag 00WT). cited at proof lines 4072-4074. Coverings induce sheaf surjections.

Source reference. Original lemma 045Y: sources/stacks-chow/tags.txt, source lines 5453–5453. Cited at proof lines 4060-4062 to read off the witnessing relation r_i; load-bearing.

Source reference. Original lemma 00WT: sources/stacks-chow/tags.txt, source lines 1195–1195. Cited at proof lines 4072-4074 for the sufficient condition; load-bearing for the last paragraph.

Lemma 0.2. *Let $\tau \in \{Zariski, étale, fppf, smooth, syntomic\}$. Let S be a scheme. Let $j : R \to U \times_S U$ be a pre-equivalence relation over S and $g : U' \to U$ a morphism of schemes over S. Let $j' : R' \to U' \times_S U'$ be the restriction of j to U'. Assume U, U', R, S are objects of a τ-site Sch_τ. The map of quotient sheaves*

$$U'/R' \longrightarrow U/R$$

is injective. If g defines a surjection $h_{U'} \to h_U$ of sheaves in the τ-topology (for example if $\{g : U' \to U\}$ is a τ-covering), then $U'/R' \to U/R$ is an isomorphism.

Proof. Suppose $\xi, \xi' \in (U'/R')(T)$ are sections which map to the same section of U/R. Then we can find a τ-covering $\mathcal{T} = \{T_i \to T\}$ of T such that $\xi|_{T_i}, \xi'|_{T_i}$ are given by $a_i, a_i' \in U'(T_i)$. By Lemma 045Y and the axioms of a site we may after refining $\mathcal{T}$ assume there exist morphisms $r_i : T_i \to R$ such that $g \circ a_i = s \circ r_i$, $g \circ a_i' = t \circ r_i$. Since by construction $R' = R \times_{U \times_S U} (U' \times_S U')$ we see that $(r_i, (a_i, a_i')) \in R'(T_i)$ and this shows that a_i and a_i' define the same section of U'/R' over T_i. By the sheaf condition this implies $\xi = \xi'$.

If $h_{U'} \to h_U$ is a surjection of sheaves, then of course $U'/R' \to U/R$ is surjective also. If $\{g : U' \to U\}$ is a τ-covering, then the map of sheaves $h_{U'} \to h_U$ is

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surjective, see Sites, Lemma 00WT. Hence $U'/R' \to U/R$ is surjective also in this case. $\square$